

MudawanSn: A Gold-Standard Wolof–Arabic Parallel Corpus for Machine Translation

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📄 hf.co/datasets/mbaye930/wolof-arabic-parallel-corpus

📄 github.com/M-mbaye30/MudawanSn

Abstract

We present MudawanSn, a gold-standard resource of 1,271 sentence-aligned pairs manually translated from Wolof into Modern Standard Arabic (MSA). The source texts are drawn from the MasakhaNER corpus (Ade-lani et al., 2022b), covering politics, society, religion, and sports in Senegalese news discourse. Although multilingual resources such as FLORES-200 (NLLB Team, 2024) and NTREX (Federmann et al., 2022) contain both Wolof and Arabic translations, to the best of our knowledge no publicly available parallel corpus has been specifically designed for the Wolof–Modern Standard Arabic language pair. We describe the corpus construction protocol, sentence alignment procedure, and quality-control workflow. We benchmark four machine translation systems spanning three architectural families: NLLB-200 (600M), mT5-base, and two AfriNLLB variants, showing that fine-tuning on MudawanSn yields substantial improvements in both translation directions. The best-performing model, AfriNLLB-12, achieves 7.76 BLEU and 30.72 chrF++ for Wolof-to-Arabic, and 8.75 BLEU and 33.08 chrF++ for Arabic-to-Wolof. The corpus is released under CC BY-NC and publicly available on Hugging Face and GitHub.

Keywords: Parallel Corpus, Low-Resource Language, Machine Translation, Wolof, Modern Standard Arabic, African NLP

1 Introduction

Wolof is the principal language of daily communication for approximately ten million speakers across Senegal, The Gambia, and Mauritania,

¹<https://github.com/masakhane-io/masakhane-ner/>

²<https://github.com/AfriNLP/AfriNLLB>

where it coexists with the respective official languages in multilingual settings (Robert, 2024). Despite this large speaker population, Wolof remains severely under-represented in natural language processing research. The few existing parallel resources pair Wolof almost exclusively with French, reflecting Senegal’s colonial and administrative context but leaving other culturally relevant language pairs entirely unaddressed.

The Wolof–Modern Standard Arabic (MSA) language pair is one such case. Senegal is a predominantly Muslim country with a long tradition of Arabic literacy through Qur’anic education, which led to the development of Wolofal, an Arabic-based Ajami writing system for Wolof (Ngom, 2017). Despite these historical and cultural ties, modern NLP resources for the Wolof–MSA language pair remain scarce. Although multilingual resources such as FLORES-200 (NLLB Team, 2024), NTREX (Federmann et al., 2022), and FineTranslations (Penedo et al., 2026) include both Wolof and Arabic, they were not specifically designed as manually curated parallel corpora for this language pair. To the best of our knowledge, no publicly available manually translated, gold-standard parallel corpus has previously been developed for Wolof–MSA machine translation. Consequently, multilingual systems such as NLLB-200 (NLLB Team, 2024) technically support both languages, yet our benchmark results indicate that their zero-shot translation quality on this language pair remains very limited.

To bridge this gap, we present MudawanSn, a publicly available Wolof–Arabic parallel corpus. Comprising 1,271 sentence-aligned pairs manually translated from Wolof into Modern Standard Arabic (MSA), our dataset is derived from Senegalese news articles in the MasakhaNER corpus (Ade-lani et al., 2022b). Our main contributions are as follows:

- The release of MudawanSn, a publicly avail-

able gold-standard Wolof–MSA parallel corpus comprising 1,271 manually translated and sentence-aligned pairs, released together with comprehensive documentation and reproduction resources.

- A detailed and reproducible construction protocol for building a gold-standard parallel corpus for a previously under-resourced language pair. The protocol encompasses sentence extraction and translation, multilingual source alignment, cross-lingual semantic verification using LASER3 (Heffernan et al., 2022), and rigorous quality-control procedures.
- A benchmark of four machine translation architectures, NLLB-200 (NLLB Team, 2024), mT5-base (Xue et al., 2021), and two AfriNLLB variants (Moslem et al., 2026), showing that fine-tuning substantially improves translation quality in both directions, with AfriNLLB-12 achieving the best overall performance (Section 4).

2 Related Work

A data paper’s contribution is best understood by situating it within the existing landscape of language resources. We therefore review (i) parallel corpora involving Wolof, (ii) multilingual resources that include both Wolof and Arabic, and (iii) the specific gap addressed by MudawanSn. Table 1 summarizes the principal resources discussed in this section.

2.1 Parallel Corpora Involving Wolof

Wolof has received increasing attention in NLP over the past few years, leading to the release of several parallel and multilingual resources. However, these resources predominantly pair Wolof with French or English, or include it as one language among many in massively multilingual collections. Table 1 summarizes the principal resources discussed below.

French–Wolof. Early work on Wolof machine translation focused almost exclusively on French. SENCORPUS (Nguer et al., 2020) was among the first French–Wolof parallel corpora developed for neural machine translation, covering multiple domains including education, religion, legislation, and society. MAFAND-MT (Adelani et al., 2022a) later introduced a publicly available, manually translated Wolof subset of 6,366 sentence pairs

collected from four Senegalese news outlets. More recently, Mbaye et al. (Mbaye et al., 2023) presented a substantially larger French–Wolof corpus of 123,000 sentence pairs for neural machine translation experiments. However, unlike MAFAND-MT, both SENCORPUS and the corpus of Mbaye et al. remain unavailable to the research community, limiting reproducibility and downstream reuse.

English–Wolof. Wolof is also represented in multilingual evaluation benchmarks. FLORES-200 (NLLB Team, 2024) provides professionally translated evaluation sets covering 200 languages, including both Wolof and Modern Standard Arabic, while NTREX-128 (Federmann et al., 2022) contains professionally translated news test sets for 128 languages. These benchmarks are intended for evaluation rather than model training and therefore do not provide large-scale parallel corpora for developing machine translation systems.

Massively multilingual resources. Recent multilingual initiatives have further expanded Wolof coverage. NLLB-200 (NLLB Team, 2024), MADLAD-400 (Kudugunta et al., 2023), SMOL (Caswell et al., 2025), and FineTranslations (Penedo et al., 2026) all include Wolof among hundreds of languages. These resources primarily target broad multilingual coverage and large-scale model pretraining rather than the development of high-quality parallel corpora for specific language pairs. In particular, FineTranslations relies largely on automatically generated translations rather than manually curated bilingual data.

What is missing. Despite these important advances, publicly available Wolof parallel resources remain overwhelmingly centered on French or English, either directly or through multilingual collections. To the best of our knowledge, no publicly available, manually translated, gold-standard parallel corpus has previously been developed specifically for the Wolof–Modern Standard Arabic language pair. MudawanSn addresses this gap by providing such a resource together with a reproducible construction protocol and machine translation baselines.

2.2 Parallel Corpora Involving Arabic and African Languages

Although Arabic is widely represented in multilingual machine translation benchmarks, dedi-

Dataset	Pair(s)	# Pairs	Pub.?	Transl.	Use	Key characteristic
SENCORPUS (Nguer et al., 2020)	Fr–Wo	~70,000	✗	Human	MT	Closed; Fr–Wo only
MAFAND-MT (Adelani et al., 2022a)	Fr/En ↔ Afr. (incl. Wo)	6,366	✓	Human	MT	No dedicated Wo–MSA pair
FLORES-200 (NLLB Team, 2024)	Multiling. (incl. Wo, Ar)	2,024	✓	Human	Eval	Evaluation benchmark only
NTREX-128 (Federmann et al., 2022)	Multiling. (incl. Wo, Ar)	1,997	✓	Human	Eval	Evaluation benchmark only
SMOL (Caswell et al., 2025)	Multiling. (incl. Wo)	~97k	✓	Human	Pretr.	English-centric focus
FineTranslations (Penedo et al., 2026)	Massively multiling. (Wo, Ar)	3.33B	✓	Auto.	Pretr.	Automatic; uncurated
Mbaye et al. (Mbaye et al., 2023)	Fr–Wo	123,000	✗	Human	MT	Closed; Fr–Wo only
MudawanSn	Wo–MSA	1,271	✓	Human	Res.+ bench.	First public, human-curated, dedicated

Table 1: Comparison of existing resources relevant to Wolof–Modern Standard Arabic machine translation. “Pub.?” indicates unrestricted public availability; “Transl.” specifies whether translations are human- or machine-generated.

cated parallel resources pairing Arabic with sub-Saharan African languages remain relatively scarce compared with English- or French-centered resources (Adelani et al., 2022a; NLLB Team, 2024). The largest multilingual collections, including OPUS (Tiedemann, 2012), NLLB-200 (NLLB Team, 2024), and MADLAD-400 (Kudugunta et al., 2023), include Arabic alongside numerous African languages. However, many African–Arabic language pairs are supported through multilingual transfer or pivoting rather than dedicated bilingual parallel corpora (NLLB Team, 2024; Kudugunta et al., 2023).

Second, recent African-focused MT projects targeting Arabic, including the ongoing AfriNLLB initiative (Moslem et al., 2026), address the resource scarcity problem at scale but do not yet provide curated, sentence-aligned, gold-standard corpora for specific African–Arabic language pairs. This remains an important limitation because multilingual transfer and zero-shot translation between low-resource language pairs generally underperform systems fine-tuned on directly parallel data (Fan et al., 2021; Adelani et al., 2022a). This limitation is also reflected in our experiments (Section 4), where multilingual models exhibit very weak zero-shot performance before fine-tuning on MudawanSn. The recent survey of Senegalese NLP (Mbaye et al., 2026) likewise identifies the lack of publicly available Wolof–Arabic resources

as an important gap for future research.

2.3 Positioning of Our Contribution

Our corpus addresses the intersection of the two gaps identified above. To the best of our knowledge, it is the first publicly available, human-curated parallel resource pairing Wolof directly with Modern Standard Arabic at the sentence level for the news domain.

MudawanSn differs from previous resources in three respects. First, it is *gold-standard*: every sentence pair has been manually translated and manually verified, in contrast to the web-mined or pivoted Arabic–African data found in massively multilingual collections. Second, it is *news-domain and culturally situated*: the source texts come from four Senegalese outlets covered by MAFAND-MT, anchoring the corpus in a register and a content universe that is directly relevant to the Wolof-speaking community. Third, it is *released with full provenance*: each pair is traceable to its upstream source through the matching procedure described in Section 3, supporting downstream studies that wish to control for source register or domain.

3 Corpus Construction and Quality Control

Building a high-quality parallel corpus for a language pair with no prior bilingual resources requires a multi-stage pipeline ensuring translation

fidelity and alignment correctness. We therefore designed and executed an end-to-end process to clean, enrich, align, and semantically verify the manually translated sentence pairs, resulting in a verified gold-standard resource. Figure 1 provides an overview of the full pipeline.

3.1 Source Data and Provenance

The Wolof source sentences were selected from the MasakhaNER corpus (Adelani et al., 2022b), which aggregates news articles from four major Senegalese online outlets: *Seneweb*, *Jotna News*, *YerimPost*, and *SocialNetLink*. These texts represent contemporary written Wolof and cover diverse thematic areas such as politics, society, religion, and sports in Senegalese public discourse.

We selected MasakhaNER as our source for three scientific reasons. First, the texts were authored by native Senegalese speakers, ensuring authentic and natural Wolof news registers. Second, the corpus is released under a CC BY-NC 4.0 license, allowing the redistribution of derivative translated resources under compatible terms. Third, the sentences had already undergone high-quality segmentation and basic preprocessing, providing a clean foundation for cross-lingual alignment. From these source texts, a subset of 1,271 sentences was selected for the translation pipeline.

3.2 Manual Translation

The Arabic portion of the corpus was translated directly from the Wolof source by a bilingual native speaker of Wolof with advanced competence in Modern Standard Arabic (MSA). The translation followed a hybrid literal-semantic strategy: literal translation was preferred whenever it yielded a grammatical and idiomatic Arabic sentence, while semantic (meaning-based) translation was applied when a literal rendering would have produced unidiomatic or ambiguous Arabic.

To ensure maximum accuracy, the translator relied on the monolingual Arabic dictionary Al-maany¹ to verify word senses and registers, and consulted with native Wolof and Arabophone speakers when translating complex cultural or idiomatic concepts. No commercial machine translation systems were used in the translation loop, avoiding any circularity bias for downstream MT evaluation.

¹<https://www.almaany.com>

3.3 Cleaning and Normalization

We designed and executed a uniform preprocessing pipeline to clean the raw bilingual texts. We applied Unicode Normalization Form C (NFC) to both the Wolof and Arabic texts to resolve decomposed diacritics and composite characters. We programmatically stripped non-standard whitespaces (e.g., non-breaking spaces) and collapsed multiple whitespace sequences into a single standard space. Punctuation marks were standardized, and any residual formatting artifacts were removed to ensure a clean, unified textual representation.

3.4 Multilingual Source Matching and Enrichment

To enrich our corpus and enable future multi-way translation studies, we designed a custom three-stage matching pipeline to pair our Wolof sentences against the larger galsenai/french-wolof-translation² dataset, thereby recovering the original French source sentences and metadata:

1. **Exact Normalized Match:** We lowercased both strings, removed punctuation, collapsed spaces, and compared them. This yielded an exact match rate of 53.89% (685 sentences).
2. **Substring Match:** For source sentences that had been segmented into multiple shorter sentence fragments, we checked if the normalized Wolof sentence was a substring of any reference sentence, recovering an additional 34.30% (436 sentences).
3. **Fuzzy Token-Overlap Match:** For sentences with minor spelling variations, we calculated the token-level intersection. A match was established if at least 85% of the words overlapped, recovering 9.99% (127 sentences).

Through this strategy, we successfully matched and enriched 98.19% (1,248 out of 1,271) of our parallel pairs with their corresponding original French source sentences (`french_source`) and source metadata (`source_dataset` like MAFAND). The remaining 1.81% are retained with MasakhaNER fallback records.

²<https://huggingface.co/datasets/galsenai/french-wolof-translation>

LASER3 cosine interval	Sentence pairs (%)
[0.0, 0.4)	0.31
[0.4, 0.6)	3.62
[0.6, 0.8)	69.47
≥ 0.8	26.59

Table 2: Distribution of LASER3 cross-lingual similarity scores over the 1,271 sentence pairs.

3.5 Sentence Alignment and Quality Control

Each of the 1,271 sentence pairs was manually inspected to ensure strict one-to-one semantic correspondence, with no omissions or translation drift.

To quantitatively validate the parallel alignment and verify translation quality, we computed cross-lingual semantic similarity scores using the LASER3 model (Heffernan et al., 2022) as our primary quantitative check of alignment quality.

The similarity scores across the 1,271 parallel sentence pairs show high parallel fidelity, yielding a mean score of 0.7508 and a median score of 0.7649. Table 2 reports the detailed distribution.

This semantic evaluation supports the robust alignment and high quality of our dataset, with 96.06% of all parallel sentence pairs scoring above 0.60 (and 26.59% scoring ≥ 0.80). All sentence pairs scoring below 0.60 were flagged and subjected to a rigorous manual review, correcting any alignment shifts, translation omissions, or spelling variations. The final corpus is released in a standard parallel format `tsv`.

Corpus Statistics. To prevent the over-fragmentation bias typical of multilingual subword tokenizers (e.g., the SentencePiece tokenizers in NLLB or mT5, which frequently over-segment low-resource languages like Wolof), all corpus statistics are reported using simple whitespace tokenization (space-delimited words). Under this space-separated count, the 1,271 sentence pairs contain a total of 25,543 Wolof words and 17,972 Arabic words, with mean sentence lengths of 20.1 words (Wolof) and 14.1 words (Arabic). The vocabulary comprises 5,321 unique Wolof word types and 8,386 Arabic word types; the larger Arabic vocabulary size reflects the morphological richness of MSA. Source texts are predominantly Senegalese news from the MAFAND collection (98.2%), with a small MasakhaNER fallback component (1.8%).

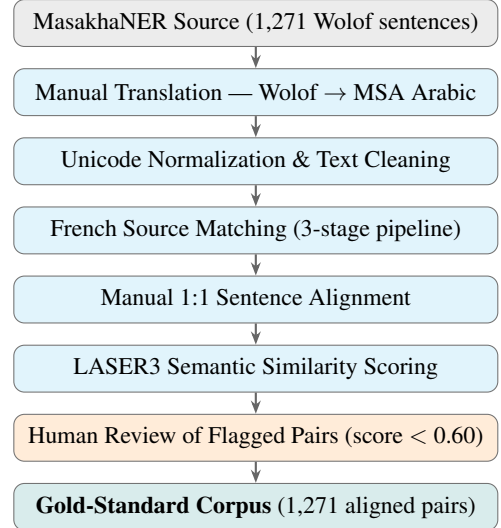


Figure 1: End-to-end pipeline from raw MasakhaNER source texts to the verified gold-standard Wolof–Arabic parallel corpus.

4 Baseline Experiments

We assess the value of the proposed corpus along two axes: the magnitude of fine-tuning gains it enables over zero-shot baselines, and the relative behaviour of model families with different degrees of African-language coverage. To this end, we evaluate four systems representing three architectural families.

4.1 Experimental Setup

We split the 1,271 sentence pairs into training (85%, $n = 1,079$), development (7.5%, $n = 96$), and test (7.5%, $n = 96$) sets. Splits are released alongside the corpus to ensure reproducible evaluation.

Models. NLLB-200 distilled 600M (NLLB Team, 2024) is a massively multilingual encoder–decoder MT model with explicit Wolof (`wol_Latn`) and Arabic (`arb_Arab`) language codes. mT5-base (Xue et al., 2021) is a generalist text-to-text model pre-trained on the multilingual mC4 corpus; we include it as an out-of-distribution lower-bound reference, since mC4 contains substantial Arabic but only marginal Wolof coverage. AfriNLLB (Moslem et al., 2026) is an African-language-specialized variant of NLLB-200. We evaluate two configurations: the full 12-encoder/12-decoder model and a pruned 8-encoder/8-decoder variant.

Training regimes. Each model is evaluated in two settings. Zero-shot: the off-the-shelf model is

Direction	Model	BLEU	chrF++	AfriCOMET _{scaled}
wo → ar	NLLB-200 (600M) zero-shot	1.12	15.42	−12.22
	NLLB-200 (600M) fine-tuned	6.47	29.31	0.19
	mT5-base zero-shot	0.04	0.17	−40.43
	mT5-base fine-tuned	0.43	11.10	−31.98
	AfriNLLB-12enc-12dec zero-shot	3.97	25.39	−28.38
	AfriNLLB-12enc-12dec fine-tuned	7.76	30.72	−24.47
	AfriNLLB-8enc-8dec zero-shot	4.53	25.53	−29.86
	AfriNLLB-8enc-8dec fine-tuned	7.05	28.40	−28.66
	NLLB-200 (600M) zero-shot	3.14	18.24	36.49
	NLLB-200 (600M) fine-tuned	7.28	30.87	41.93
ar → wo	mT5-base zero-shot	0.02	2.57	−41.11
	mT5-base fine-tuned	0.09	2.50	−50.83
	AfriNLLB-12enc-12dec zero-shot	3.08	21.09	−14.71
	AfriNLLB-12enc-12dec fine-tuned	8.75	33.08	−3.18
	AfriNLLB-8enc-8dec zero-shot	5.66	26.31	−7.49
	AfriNLLB-8enc-8dec fine-tuned	8.18	31.95	−1.53

Table 3: Baseline machine translation results on the test split. Best score per direction and metric in bold.

used directly on the test set in both directions. Fine-tuned: we perform full-parameter bidirectional fine-tuning, jointly optimizing wo→ar and ar→wo, for 10 epochs.

Metrics. We report BLEU (Papineni et al., 2002) and chrF++ (Popović, 2017) computed with sacreBLEU (Post, 2018), and AfriCOMET (Wang et al., 2024), an African-centric neural metric built on the AfroXLM-R encoder. chrF++ is included because character-level metrics correlate better with human judgement on morphologically rich languages such as Wolof and Arabic.

4.2 Results

Table 3 reports the full benchmark.

In the wo→ar direction, all multilingual MT models benefit consistently from fine-tuning. NLLB-200 improves by +5.35 BLEU and +13.89 chrF++. The African-specialized AfriNLLB-12 reaches the highest lexical scores (7.76 BLEU, 30.72 chrF++), with the pruned AfriNLLB-8 within 0.7 BLEU of the full model. The same pattern holds in ar→wo: NLLB-200 improves from 3.14 to 7.28 BLEU and AfriNLLB-12 reaches the best lexical performance at 8.75 BLEU and 33.08 chrF++.

Two observations stand out. First, mT5-base behaves asymmetrically: fine-tuning yields a small but measurable gain when Arabic is the target (chrF++ rises from 0.17 to 11.10), but no gain when Wolof is the target. This is consistent with mT5’s pre-training mix—mC4 contains substan-

tial Arabic but virtually no Wolof—and suggests that the corpus is a useful fine-tuning resource only for models that already have some Wolof coverage. NLLB-200 and AfriNLLB satisfy this condition; mT5-base does not, despite seeing the same training data.

Second, lexical and neural metrics do not fully agree on system ranking. AfriNLLB-12 obtains the best BLEU and chrF++ scores in both directions, while NLLB-200 fine-tuned obtains the highest AfriCOMET scores. Since the Wolof–Arabic pair is not covered by AfriCOMET’s training data, we report these scores for completeness but base our comparison primarily on the lexical metrics, leaving a human evaluation of system outputs to future work. These results demonstrate that MudawanSn provides a useful supervision signal for multilingual models that already possess some degree of Wolof representation.

5 Conclusion

We have introduced a publicly available Wolof–Arabic parallel corpus, a gold-standard resource of 1,271 manually translated and aligned sentence pairs drawn from Senegalese news sources. We documented the construction protocol and reported baseline machine translation experiments spanning three architectural families. Fine-tuning on our corpus produces consistent improvements over zero-shot baselines for all multilingual MT models, with African-specialized variants reaching the highest scores. Beyond machine translation, the corpus of-

fers a foundation for contrastive linguistic studies of Wolof and Arabic, for the development of Wolof-aware tokenizers and morphological analyzers, and for broader NLP research on low-resource African language pairs involving Arabic. By releasing the corpus, splits, and code openly under a CC BY-NC license, we aim to open a line of work that has so far been entirely absent from the computational linguistics literature.

Limitations

Script coverage. Our corpus is limited to the Latin orthography of Wolof and therefore does not include Wolofal. Given the scarcity of publicly available Ajami-Wolof resources and Latin–Ajami transliteration tools, extending MudawanSn to cover Wolofal remains an important direction for future work. Such an extension would enable the study of the impact of shared Arabic-derived scripts on Arabic–Wolof machine translation.

Corpus size. At 1,271 sentence-aligned pairs, the corpus is modest in absolute size. This reflects the manual effort and multi-stage quality control required to construct a gold-standard bilingual resource from scratch for a language pair with no prior parallel data. Scaling the corpus up and expanding it to additional domains is our primary direction for future work.

Single-translator design. The entire Arabic side was produced by a single bilingual translator. Although the workflow included dictionary-based verification, consultation with native speakers for culturally loaded items, and a semantic verification stage with human review of flagged pairs, no independent second translation or inter-translator agreement measure was collected. Translation choices therefore reflect one translator’s stylistic and lexical preferences, particularly in register selection within Modern Standard Arabic.

Coverage of idiomatic expressions. During translation, highly localized or culture-specific Wolof idiomatic expressions were deliberately avoided in favour of constructions that admit a precise mapping to Modern Standard Arabic. This choice was made to ensure unambiguous parallel quality and alignment robustness for machine translation training, but it inevitably underrepresents the figurative register of Senegalese news Wolof.

Domain coverage. All source texts come from Senegalese online news outlets, with politics and civic discourse over-represented relative to other registers. Conversational, literary, religious, and technical domains are entirely absent. Downstream systems fine-tuned solely on this corpus will inherit this bias and should be expected to transfer imperfectly to other registers.

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